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Reconstruction of highly porous structures from FIB-SEM using a deep neural network trained on synthetic images

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Summary

Combining scanning electron microscopy with serial slicing by a focused ion beam yields spatial image data of materials structures at the nanometer scale. However, the depth of field of the scanning electron microscopic images causes unwanted effects when highly porous structures are imaged. Proper spatial reconstruction of such porous structures from the stack of microscopic images is a tough and in general yet unsolved segmentation problem. Recently, machine learning methods have proven to yield solutions to a variety of image segmentation problems. However, their use is hindered by the need of large amounts of annotated data in the training phase. Here, we therefore replace annotated real image data by simulated image stacks of synthetic structures – realizations of stochastic germ–grain models and random packings. This strategy yields the annotations for free, but shifts the effort to choosing appropriate stochastic geometry models and generating sufficiently realistic scanning electron microscopic images.

Introduction

Physical properties of materials like mechanical strength, thermal or electrical conductivity or filtration efficiency depend strongly on microstructural features like volume fractions of phases, wall or strut thickness distributions for foams or sponges, length and orientation distributions of reinforcing fibres in composites or size and shape distributions of cells or pores in cellular or porous media. Microscopic images have been used to gain deeper insight into materials microstructures since nearly 200 years. At about the same time, Delesse (1847) deduced the 3D volume fraction from the 2D area fraction (Baddeley & Vedel Jensen, 2005). Since then, a wide variety of sophisticated methods have been developed to gain 3D geometric information based on 2D image data, see, e.g. Baddeley & Vedel Jensen (2005). Researchers have tried to obtain spatial information by serial sectioning for at least 100 years, too (see Uchic, 2011 referring to Forsman, 1918).

During the last decades, with the rise of microcomputed tomography, 3D imaging has finally superseded the classical approach in many applications and enabled various new analysis methods. With materials structures getting more and more tailor-suited to specific needs and nanostructuring becoming more widely accessible, the demand for 3D imaging well beyond the micrometer scale grew, too. Serial slicing by a focused ion beam (FIB) combined with scanning electron microscopy (SEM) of the slices as first described by Inkson *et al.* (2001) and Holzer *et al.* (2004) fulfills this demand delivering high-quality 3D images of materials structures at the scale of 3–200 nm.

The typically very large focal depth of the SEM images causes the so-called shine through artefacts (Prill & Schladitz, 2013) when imaging highly porous structures. That is, the solid structure from deeper layers is visible through the pores. Thus, the stack does not represent the true 3D structure. Reconstructing the proper spatial structure is in general not yet resolved despite of several algorithms developed particularly for this purpose (Prill *et al.*, 2013, 2020; Salzer *et al.*, 2014, 2015; Holzer *et al.*, 2016; Penner & Holzer, 2018; Ott *et al.*, 2019). All these solutions are tailored to specific structures and SEM modes and therefore not generally applicable.

Machine learning methods have recently solved segmentation tasks where classical methods failed so far. Röding *et al.* (2019) segment 3D imaging by serial slicing using an FIB and imaging the slices by SEM (FIB-SEM) images using a random forest. Convolutional neural networks (CNNs) are readily available. Çiçek *et al.* (2016) and Qi *et al.* (2017) have used them successfully also for 3D image segmentation. However, the typical need of CNNs for large training data forms a serious obstacle in our setting. FIB-SEM is still a rather elaborate imaging method, in particular for highly porous structures. Moreover, manual annotation, that is, interactive segmentation, is time-consuming and error-prone, again due

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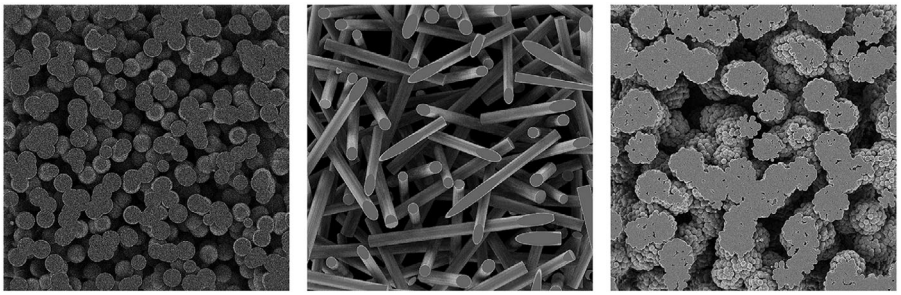


Fig. 1. Examples of the backscattered electron signal of synthetic scanning electron microscopy image data of realizations of stochastic geometry models. From left to right: Boolean model of spheres, random packing of cylinders and Cox–Boolean model of spheres.

to the shine through artefacts that are hard to distinguish from the true foreground even for humans.

Stacks of simulated SEM images of synthetic structures like the ones shown in Figure 1 could resolve both these problems. First, the ground truth is readily available as direct discretization of the synthetic structures obtained as realizations of stochastic geometry models. Second, using efficient simulation techniques as devised by Prill & Schladitz (2013), generating the FIB-SEM image data from these synthetic structures is much cheaper than real experiments. The simulated FIB-SEM stacks together with the ground truth can then be used to train the CNN. First attempts reported by Fend *et al.* (2020) and based on a network architecture close to U-net 3D by Çiçek *et al.* were promising. Here, further progress is described and the trained CNN is applied to real data.

Synthetic FIB-SEM image data

In order to cover a wide range of local textures as well as spatial distributions of pores and solid material, we use a variety of stochastic geometry models (Ohser & Schladitz, 2009; Altendorf & Jeulin, 2011; Redenbach *et al.*, 2014). For the fast simulation of FIB-SEM image stacks by the method of Prill & Schladitz (2013), an analytic description of their realizations is needed, e.g. as a list of centres and radii. Thus, germ–grain models are particularly suitable. Here, we use Boolean models (Stoyan *et al.*, 1995; Schneider & Weil, 2008) of spheres and straight circular cylinders as did Salzer *et al.* (2015) and a Cox–Boolean model of small spheres nested in large ones (Jeulin, 2012). These are complemented by force biased packings of spheres (Bezrukov *et al.*, 2001, 2002), random sequential adsorption packings of straight circular cylinders and curved fibre systems of Altendorf & Jeulin (2011). Realizations are both stored as lists of geometric objects and discretized to 3D images.

Synthetic FIB-SEM stacks are simulated from the model realizations using the method described by Prill & Schladitz (2013). Virtual slices are calculated. The actual SEM is simulated based on electron paths computed by MONSEL II by Lowney (1996). Several acceleration techniques are applied to speed up the simulation such that it produces the desired 200–900 images in practicable time. Electron paths are precomputed and randomly chosen, paths are simplified where no interaction is possible, secondary electron generation is cut off when the escape depth is reached.

Here, the backscattered electron (BSE) signal is used, the solid component is assumed to be carbon, and the dwelling time is 1 µs. The primary electrons have an energy of 5 keV as chosen by Prill & Schladitz (2013). For details on SEM parameters, we refer to Goldstein *et al.* (2003). For the sake of variety, FIB-SEM stacks with deviations in one of the imaging parameters are taken into account, too. The deviating parameter is included in the image's name in these cases. Parameters of the models as well as the imaging parameters are given in Table 1.

Throughout, the SEM images are parallel to the xy-plane whereas the z-direction is the one they are stacked in.

Network setup

In the following, we use a CNN to solve our segmentation task. We first summarize the main ideas behind CNNs and clarify basic concepts.

A classical neural network is a biologically inspired method of assembling and linking a series of 'neurons' in several layers. Each neuron receives a vector of features, x, as an input and computes a scalar output value as a weighted average of this input. The weights w are the parameters of this particular neuron. In a fully connected network, successive layers are linked by feeding the vector of all outputs of the preceding layer as input to the neurons of the succeeding layer. Roughly speaking, the goal is to find weights w for all neurons such that the value of the final output $\hat{y}$ is close enough to the 'ideal value' or simply the ground truth y, where distance is measured by a loss function.

Nonlinear networks are a generalization offering more flexibility. They relate the neurons to the output by activation functions. Currently popular activation functions are, e.g. sigmoid, tanh and rectified linear unit. The latter leaves the input value untouched if it is positive and returns zero otherwise. Soft-max activation scales feature maps to yield probabilities by assigning each feature its exponential and normalizing by the sum of

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Table 1. Parameters of the geometry models and the SEM simulation. Solid structure is assumed to be carbon, geometry models are stationary and isotropic if not stated otherwise.

Model type	Name	$1 - V_V$ [%]	Typical grain [px]	Image size [px]	SEM mode	Dwell time [μs]
Boolean models						
Spheres	BoolSpheres	60	$r \equiv 15$	600×600×600	BSE	1
Spheres	BoolSpheres0p1μs	60	$r \equiv 15$	600×600×600	BSE	0.1
Spheres	BoolSpheres5μs	60	$r \equiv 15$	600×600×600	BSE	5
Spheres	BoolSpheres20px[a]	60	$r \equiv 20$	600×600×600	BSE	1
Spheres	BoolSpheres35px	60	$r \equiv 35$	600×600×600	BSE	1
Spheres	BoolSpheresSE	60	$r \equiv 15$	600×600×600	SE	1
Spheres	BoolSpheresNon[b]	73	$r \equiv 15$	600×600×600	BSE	1
Cylinders	BoolCyl	68	$r \sim \text{Unif}[10, 15]$ $\ell \sim \text{Unif}[100, 220]$	425×425×425	BSE	1
Cox–Boolean models						
Spheres	CoxBool[a]	60	$r \equiv 6,\ R \equiv 40$[c]	600×600×598	BSE	1
Spheres	CoxBool15	30	$r \equiv 15,\ R \equiv 20$[c]	850×850×1000	BSE	1
Random packings						
Spheres	FB[d]	40	$r \sim \text{LogNorm}(3.96, 0.02)$	900×900×471	BSE	1
Force biased						
Cylinders	RSACyl[a,e]	88	$r \sim \text{Unif}[10, 15]$	600×600×300	BSE	1
RSA			$\ell \equiv 200$			
fibres	AJ	86	$r \equiv 9, \ell \equiv 100$	200×200×400	BSE	0.1
Altendorf–Jeulin			$\kappa_1 = 10,\ \kappa_2 = 100$			

[a]Visualized in Figure 1.
[b]Nonstationary, intensity a linear function of the y-coordinate.
[c]Spheres of radius r nested in spheres of radius R.
[d]Mean radius 53, coefficient of variation 0.02.
[e]Anisotropic, orientation distributed according to the special case of the angular central Gaussian distribution used by Schladitz *et al.* (2006) with parameter $\beta = 0.9$ shifting the distribution slightly towards the z-direction.
BSE - backscattered electron signal, SE - secondary electron signal, RSA - random sequential adsorption, Unif - uniform distribution, LogNorm - log normal distribution

all these exponentials (see Ronneberger *et al.*, 2015, section 3). For more information on activation functions and their effect on the whole deep neural network, we refer to Glorot & Bengio (2010).

A CNN architecture is formed by a sequence of layers of varying types like convolution or pooling layers. All layers, except input and output layers, consist of a set of neurons with the trainable weights. In a convolutional layer, linear filters are applied across width and height of the input while extending through the entire depth. The convolution is given by the dot product between the learnable entries of the filter and the input values at every position. The output of the layer is the so-called feature map with all computed dot products. The number of filters applied in each layer as well as their support sizes, are chosen during the design of the network. Convolutional layers are often succeeded by batch normalization layers scaling the values of the feature map such that they are standardized in the statistical sense as described, e.g. by Ioffe & Szegedy (2015). That is, the sample mean is subtracted and all values are divided by the sample standard deviation.

Pooling layers are introduced to reduce the size of every feature map through downsampling using nonlinear functions. A reduction of the spatial representation of the features decreases the total number of weights and computations needed in the network. A widely used pooling function is max-pooling which partitions the input into a set of nonoverlapping subregions. Subsequently, for each subregion, it returns the maximum.

The training aims at iteratively improving the weights of all filters such that the error measured by the loss function is minimized. To this end, the weights of the net are optimized in each iteration to match a part of the training data. Due to restricted computational resources, not all training data can be processed in the same step. Instead, random subsets of the data called batches are used. Another prechosen part of the training data is reserved for validation.

Commonly used first-order optimization methods are stochastic gradient descent as used, e.g. by Robbins & Monro (1951), accelerated gradient descent applied, e.g. by Nesterov (2018), the Adam optimizer using adaptive estimates of lower-order moments as introduced by Kingma & Ba (2015)

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and adaptive learning rate methods like AdaGrad introduced by Duchi *et al.* (2011) and improved to AdaDelta by Zeiler (2012). The step width of the gradient descent and related methods is controlled by a small, positive number – the so-called learning rate. The smaller the learning rate, the longer the network has to be trained. Moreover, if the learning rate is chosen too small, the network might get stuck in a suboptimal state. However, if chosen too large, the true optimum might be overseen. For a survey of optimization methods including high-order approaches from a machine learning perspective, see Sun *et al.* (2019).

The full training process consists of so-called epochs: In each epoch, a random subset of the training data is used for improving the weights. Batches of data for training are taken from this subset. An epoch ends after a prescribed number of optimization steps or when all data from the predefined subset has been used at least once. Then, success is measured based on the reserved validation data. For the next epoch, new subsets are chosen for both, the actual training and the validation. A typical choice for the number of epochs is 100. If the training is extended for too many epochs, there is the danger of overfitting the weights to the training data which would result in bad prediction for new, unknown test data. An indicator for this is much higher accuracy on the current training subset than for the validation set.

For practical reasons, available training data typically represent only a rather small fraction of the input domain. This is usually counterbalanced by augmentation of the training data. That is, training data are transformed in order to increase their amount and to be more representative.

To prevent overfitting, a regularization technique is used: After a certain layer, a prescribed proportion of neurons is randomly chosen to 'drop out'. That is, these neurons are ignored in the following by setting their output to a noninformative value like zero in the case of rectified linear unit activation. For more details about this dropout technique, see, Gal & Ghahramani (2016).

In the following section, we explain the specific network architecture used in this study.

Network architecture

The CNN we use is the 3D U-net as specified by Çiçek *et al.* (2016), which is in turn an adaption of the original U-net by Ronneberger *et al.* (2015). U-net consists of two paths, where the first one – the encoder – is used to capture the context of the image. It executes alternating convolutional and pooling layers for downsampling the feature map sizes while simultaneously increasing their number. Once the thinnest layer – the bottleneck of the network – is passed, the symmetric second path expands the feature maps again. This gives the network architecture the characteristic U-shape the name was derived from.

The U-net encoder resembles a classic CNN with all convolutional layers being activated with rectified linear unit. They are followed by batch normalization and alternating with pooling layers. In the convolution layers, linear filters with mask size $3 \times 3 \times 3$ voxels are applied in each voxel, that is we apply stride $1 \times 1 \times 1$. Features are pooled in $2 \times 2 \times 2$ voxel cubes that are moved with stride $2 \times 2 \times 2$ voxels. In the bottleneck, a dropout layer with probability 0.5 is used as indirect augmentation. The expanding path enables precise localization in the original image dimensions by using upsampling layers instead of pooling layers and concatenating the output with the corresponding step in the encoder. Finally, one additional convolutional layer with soft-max activation is applied to yield probabilities of class membership.

The Adam optimizer as introduced by Kingma & Ba (2015) is used throughout. Figure 2 contains a sketch of our architecture including numbers of layers.

Loss function

We train the network with a weighted cross-entropy loss function. The classical cross-entropy by Rubinstein (1997) is widely used in machine learning to measure the similarity between the ground truth and its prediction in the binary classification task.

Ronneberger *et al.* (2015) refine this loss by introducing pixel-wise weights which reinforce the weighting of specific image regions in the loss calculation. That is, the weights force the network to learn an accurate classification in these regions since they are more important in the segmentation task. This is achieved by so-called weight maps for the training images.

Weight maps

A weight map $\omega : \Omega \to \mathbb{R}^+,\ \omega(x) = \omega_x$, yields for every voxel $x \in \Omega \subset \mathbb{Z}^3$ its priority in the loss calculation. With this definition, the (binary) weighted cross-entropy (WCE) as used by Ronneberger *et al.* (2015) is

$$\mathrm{WCE}(\hat{y}_{\mathrm{SM}}, y) = -\sum_{x\in\Omega} \omega(x)\big((1-y^x)\log(1-\hat{y}^x_{\mathrm{SM}}) + y^x \log(\hat{y}^x_{\mathrm{SM}})\big), \quad (1)$$

where $\hat{y}^x_{\mathrm{SM}} \in [0, 1]$ represents the soft-max prediction of the value of voxel x, i.e. the probability of being a foreground voxel, and $y^x \in \{0, 1\}$ is the value of voxel x in the ground truth. The loss thus penalizes at each position the deviation from the ground truth value with the factor defined in the weight map.

The weight map of Ronneberger *et al.* (2015) is motivated by their biomedical cell segmentation task. Their goal was to label all visible cells and to separate touching cells by small borders. The weight map therefore emphasizes pixels close to more than one label and thus promotes the necessary

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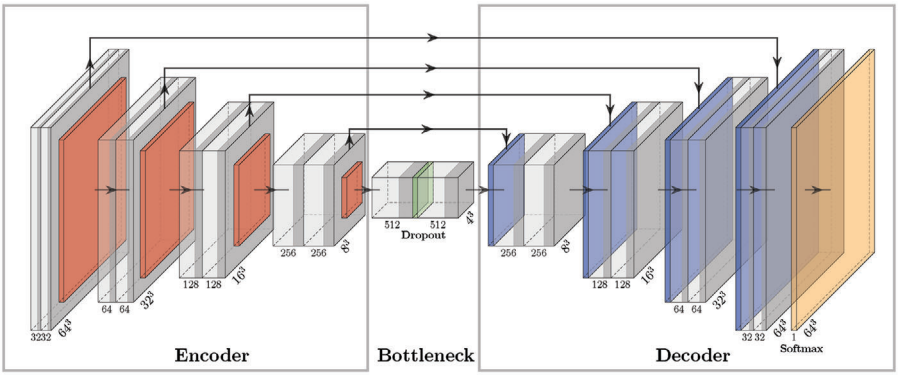


Fig. 2. Applied U-net architecture, sketch created using PlotNeuralNet by Iqbal (2018). The grey boxes represent convolutional layers followed by batch normalization and being activated with rectified linear unit. Max-pooling layers are red and the corresponding upsampling layers blue. The final output layer is a convolutional layer with soft-max activation (yellow). During training, dropout is used in the bottleneck of the network (green layer). Numbers of layers are given below the corresponding cuboidal grey boxes. Output sizes of pooling layers and input sizes for upsampling layers are given by numbers on the side of the grey boxes.

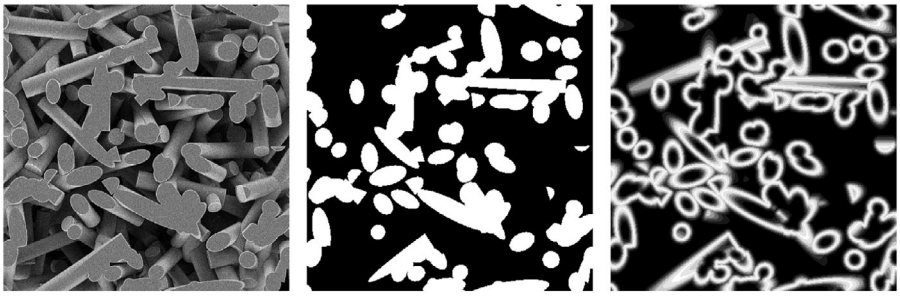


Fig. 3. Example of the calculated weight map for a realization of a Boolean model of cylinders. From left to right: simulated backscattered electron signal of a model realization, the true foreground in the front plane and the corresponding weight map.

separation of touching objects. Even though this is not our task, the precise segmentation of the surface of the solid constituent is crucial for correct reconstruction. We therefore use a weight map emphasizing voxels close to the solid constituent's surface. It is precalculated for the training volume using the Euclidean distance transform and the formula

$$\omega : \Omega \to \mathbb{N},\ \omega(x) = \lfloor 1 + \omega_0 \exp\left(-2d^2(x)/\sigma\right) \rfloor, \qquad (2)$$

where $d : \Omega \to \mathbb{R}^+$ is the distance to the closest border voxel measured in voxel edge lengths and $\lfloor \cdot \rfloor$ denotes the floor function. This definition ensures integer valued weights, that deviate from 1 only for voxels very close to the interface between solid constituent and pore space. In the following, we use $\omega_0 = 10$ and $\sigma = 4$ similar to the parameters chosen by Ronneberger *et al.* (2015). The weight map of the training volume is visualized in Figure 3.

Evaluation metrics

Frequently, in segmentation tasks, the class sizes vary strongly. A bad choice for the evaluation metric can lead to misinterpreted results in these cases. We follow Taha & Hanbury (2015) in using the dice coefficient (DC) based on the spatial overlap of ground truth and prediction. In order to define this popular metric and further quality measures for the prediction, denote by $y : \Omega \to \{0, 1\}$ the ground truth and by $\hat{y} : \Omega \to \{0, 1\}$ the binary predicted image derived by global thresholding on the soft-max prediction and with foreground voxel value 1 and background voxel value 0. Now let

$$\mathrm{TP} = \sum_{x \in \Omega} \hat{y}^x y^x, \qquad (3)$$

$$\mathrm{TN} = \sum_{x \in \Omega} (1 - \hat{y}^x)(1 - y^x), \qquad (4)$$

$$\mathrm{FP} = \sum_{x \in \Omega} (1 - \hat{y}^x) y^x, \quad \text{and} \qquad (5)$$

$$\mathrm{FN} = \sum_{x \in \Omega} \hat{y}^x (1 - y^x) \qquad (6)$$

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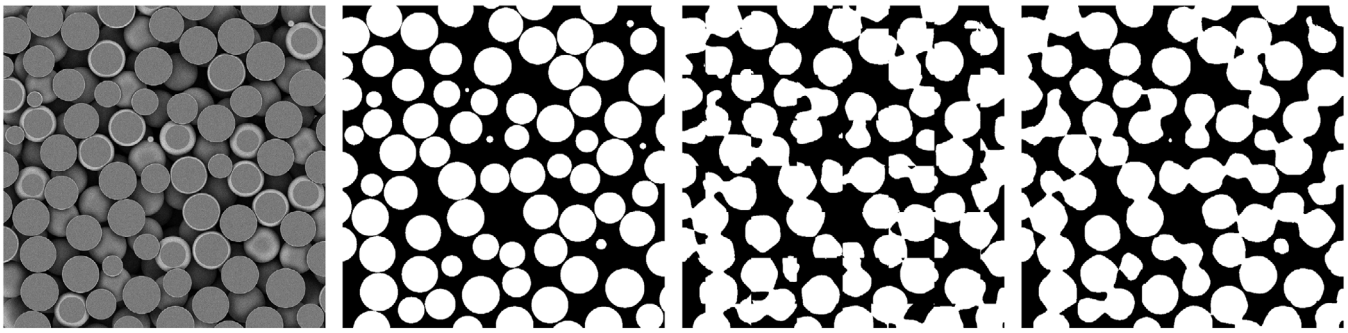


Fig. 4. Example illustrating the improvements gained by tiling with overlap. From left to right: backscattered electron signal of force biased packing of spheres, the corresponding ground truth, prediction obtained with no overlap and prediction obtained with 40 px overlap.

be the cardinalities of the sets of true positive, true negative, false positive and false negative voxels in the prediction. The dice coefficient is defined as

$$\mathrm{DC}(\hat{y}, y) = \frac{2\sum_{x\in\Omega}\hat{y}^x y^x}{\sum_{x\in\Omega}\hat{y}^x + \sum_{x\in\Omega} y^x} = \frac{2\mathrm{TP}}{\mathrm{FN} + 2\mathrm{TP} + \mathrm{FP}} \in [0, 1]. \quad (7)$$

The prediction is said to be correct if $\mathrm{DC}(\hat{y}, y)$ is close to 1.

In addition, we evaluate the false negative rate (FNR) and the false positive rate (FPR, also called fallout) since those two metrics and DC are pairwise nearly uncorrelated. The FNR indicates an underestimation of the volume fraction whereas an increased FPR will show problems due to gloss effects and shine through artefacts in the FIB-SEM images. Both metrics can be calculated with the cardinalities TP, TN, FP and FN in the following way:

$$\mathrm{FNR}(\hat{y}, y) = \frac{\mathrm{FN}}{\mathrm{FN} + \mathrm{TP}}, \quad \mathrm{FPR}(\hat{y}, y) = \frac{\mathrm{FP}}{\mathrm{FP} + \mathrm{TN}}. \quad (8)$$

Tiling

Computational feasibility enforces the use of rather small 3D patches (64^3 voxels) compared to the total image sizes (500^3–600^3 voxels) in the training process.

Simple tiling and assembling leads to strong boundary artefacts, since the segmentation in the border regions is less accurate. We therefore use a sliding window technique with overlapping regions. The regions can overlap up to 40 voxels, depending on the local structure sizes. In these overlapping regions, we use the voxel-wise mean of all predictions that contain the voxel. To correct for missing voxels at the image borders, patches are filled by reflecting at the image borders.

The resulting predicted image is still a grey value image due to using the mean of all predictions. Hence, an additional threshold, usually in [0.5, 0.75], is used to transform the output into a binary segmentation. Figure 4 shows exemplary the improvements that have been obtained with this tiling strategy.

Data augmentation

Ronneberger *et al.* (2015) augment their training data excessively. In particular, they deform the images elastically, to force the network to learn invariance with respect to transformations of this type. More precisely, elastic distortion as described by Simard *et al.* (2003) is applied: Offsets for the x- and y-components of each pixel are drawn independently identically uniformly distributed in $[-1, 1]$. The resulting spatial displacement field is subsequently convolved with a 3D Gaussian filter with standard deviation 10. Finally, the pixel values of each slice are interpolated bi-cubically using these displacement vectors.

As Ronneberger *et al.* (2015) who used only 2D image data and Çiçek *et al.* (2016) who could dispose of 2D training data, only, we leave the z-components untouched. Being the slicing direction, the z-direction is singled out in our case and therefore we apply the elastic transformation to each individual slice. In real FIB-SEM data, it is affected not only by the shine through artefacts, but also possible misalignments of the SEM images in the stack and the typical coarser resolution due to the FIB slicing being coarser than the SEM resolution.

We combine the elastic deformations with brightening or darkening the images by spreading the grey values according to a power law with positive or negative exponent and rotating or transposing the 3D patches to use any of the cube's isometries similar to Çiçek *et al.* (2016).

Transformations are applied randomly to create arbitrary combinations and thus achieve maximum possible variation in the training data. To ensure that the training data represent various sizes of the local structures, the patches are selected with a random crop and resize approach, where the scale factor varies, too. That is, smaller (cropped) patches of randomly chosen size are extracted and rescaled to the input dimensions using bilinear interpolation.

Training

We train the network in epochs consisting of 50 steps and using batches of four 64^3 voxel image patches. This choice

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Table 2. Results of networks trained using either the Boolean cylinder model (BoolCyl) or the Boolean spheres model (BoolSpheres). 'Augmentation' indicates that the data have been augmented as proposed during training but no weight maps have been applied. 'Weight maps' means that weight maps have been used but no augmentation.

Model Training\Test		BoolCyl (fibres)	BoolSpheres (spheres)	RSACyl (fibres)	CoxBool (spheres)	AJ (fibres)	FB (spheres)
BoolSpheres	DC	0.36	0.90	0.77	0.29	0.76	0.98
Augmentation only	FNR	0.68	0.17	0.36	0.80	0.67	0.026
	FPR	0.2	0.0004	0.003	0.1	0.003	0.007
BoolSpheres	DC	0.96	0.98	0.97	0.97	0.84	0.98
Weight maps only	FNR	0.06	0.03	0.04	0.06	0.27	0.02
	FPR	0.002	0.00001	0.003	0.0008	0.0001	0.01
BoolCyl	DC	0.99	0.99	0.77[a]	0.95	0.65	0.97
Weight maps only	FNR	0.01	0.008	0.28	0.09	0.29	0.01
	FPR	0.003	0.0004	0.02	0.0006	0.14	0.06

[a]Rendering of result in rightmost subfigure of Figure 8.
DC - dice coefficient, FPR - false positive rate, FNR - false negative rate

Table 3. Results of the network trained using solely the Boolean cylinder model (BoolCyl) with the proposed augmentation tools and weight maps.

	BoolCyl (fibres)	BoolSpheres[a] (spheres)	RSACyl[a] (fibres)	CoxBool[a,b] (spheres)	AJ (fibres)	FB (spheres)
DC	0.95	0.95	0.96	0.96	0.95	0.99
FNR	0.08	0.09	0.005	0.06	0.05	0.01
FPR	0.003	0.0001	0.01	0.12	0.007	0.03
$\bar{\ell}_x/\bar{\ell}_z$	0.98	1.02	0.80	0.94	0.99	0.91

	BoolSpheres 0p1μs	BoolSpheres 5μs	BoolSpheres 20px	BoolSpheres 35px	BoolSpheres SE	BoolSpheres Non[c]
DC	0.91	0.99	0.99	0.97	0.94	0.97
FNR	0.03	0.01	0.003	0.003	0.006	0.06
FPR	0.12	0.002	0.008	0.03	0.1	0.002
$\bar{\ell}_x/\bar{\ell}_z$	0.72	1.01	0.99	0.89	0.77	1.01

[a]Result for one slice visualized in Figure 6.
[b]Volume rendering of result shown in Figure 5.
[c]Volume rendering of result shown in Figure 9.
DC - dice coefficient, FPR - false positive rate, FNR - false negative rate

is a trade-off between capturing enough structural information in a patch and limiting the data to be stored in the RAM at the same time. With the architecture shown in Figure 2, we have to optimize approximately 21 million weights in each step. Doubling the number of filters in the first layer would result in four times as many parameters.

To find out which stochastic geometry models are most suitable as training data, we trained two networks that only differ in the geometry model used in training, namely, the Boolean model of spheres and the Boolean model of cylinders. We divided both image volumes into disjoint training, validation and test parts. For the training, we used the weight maps but did not apply any data augmentation. Each network was then trained for 100 epochs. The initial learning rate of 10^{-3} is halved every 10 epochs. After 100 epochs, the training showed convergence in terms of the validation loss.

As a second step, we tested the influence of the augmentation and the weight maps by comparing one network using the augmentation techniques described above (except not using any resizing of the patches) but no weight maps with the networks described above that use weight maps but no augmentation. The results are summarized in Table 2. For the model trained without weight maps, the weighted cross-entropy (1) is not applicable. Thus $1 - \mathrm{DC}(\hat{y}, y)$ was used as loss function in this case.

The results of a third network using both, augmentation and weight maps, are reported in Table 3 as they turned out

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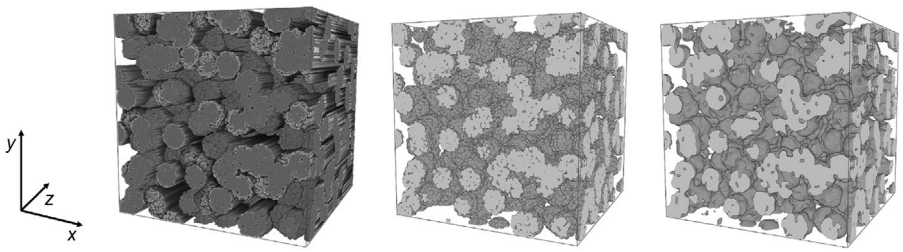


Fig. 5. Results for the Cox–Boolean model realization. Left: stack of synthetic backscattered electron images. Centre: ground truth. Right: reconstruction result of the trained convolutional neural networks. The accuracy of the segmentation is dice coefficient 0.96, false negative rate 0.06 and false positive rate 0.12.

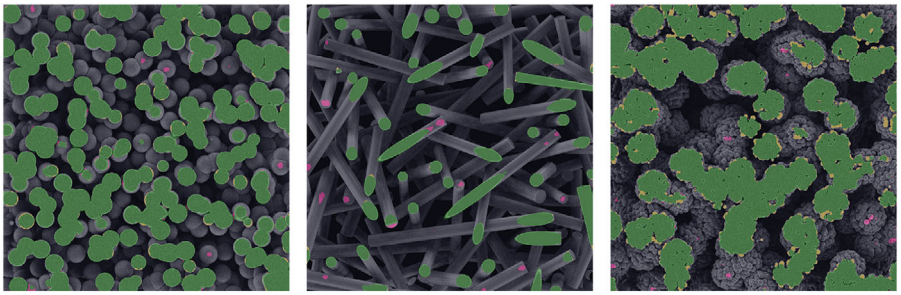


Fig. 6. Synthetic scanning electron microscopy image data overlaid with segmentation results for the examples featured in Figure 1. From left to right: BoolSpheres, RSACyl and CoxBool. Green voxels indicate correctly classified foreground voxels (TP), yellow wrongly classified as background (FN) and pink wrongly classified as foreground (FP).

to yield the best segmentation in general. Table 2 emphasizes clearly the essential impact of the weight maps on the accuracy of the estimation. They improve the segmentation significantly compared to using solely data augmentation during training. Comparing the two models that have been trained with weight maps we observe similar results for the Boolean models and the two other sphere based models (FB, CoxBool).

In case of the Altendorf–Jeulin and the random sequential adsorption cylinders realizations, the network trained on the BoolCyl volume resulted in high FNRs, showing that large parts of the structure were not found. Further inspection of the RSACyl segmentation revealed that the result was as good as for the other model in most of the volume but in some parts (entire slices in *xy*-plane) almost no foreground was correctly detected. These black slices yield the high FNR. This could be due to the extremely high porosities (90% and 86% for RSACyl and Altendorf–Jeulin, respectively, see Table 1) combined with our tiling. Moreover, the Altendorf–Jeulin data set features much larger local pores as visible in the rightmost subfigure of Figure 7. For an illustration of the impact of a rather low dice coefficient of 0.77, we refer to the rightmost subfigure of Figure 8 showing a rendering of the reconstruction obtained by training with weight maps but without augmentation.

Results

Synthetic data sets

Despite of the findings reported in Table 2, we use a network trained on the Boolean cylinder data set and not on the Boolean spheres model in the following since this network performed best in preliminary tests on FIB-SEM images of real structures. The training image BoolCyl represents one realization of the corresponding geometry model with pore volume fraction 68%. The minimum length of the cylinders equals 100 voxels and the maximum length 200 voxels. Only the BSE signal is used in the learning process. The network uses the weight map emphasizing the edges and training data is augmented as described above including random resizing.

Table 3 reports the results obtained with this network for the synthetic data listed in Table 1. Overall, the segmentations are very accurate for the chosen test volumes representing a wide variety of geometries and imaging parameters. See also Figures 5,6 for visualizations of results.

The CoxBool image features a very finely structured, rough surface (see Fig. 5). Consequently, the FPR is increased as the network has not seen this type of structure during training. For the two images BoolSpheres0p1μs and BoolSpheresSE, the FPR is increased because the grey value distributions of these two data sets deviate strongly from those of the

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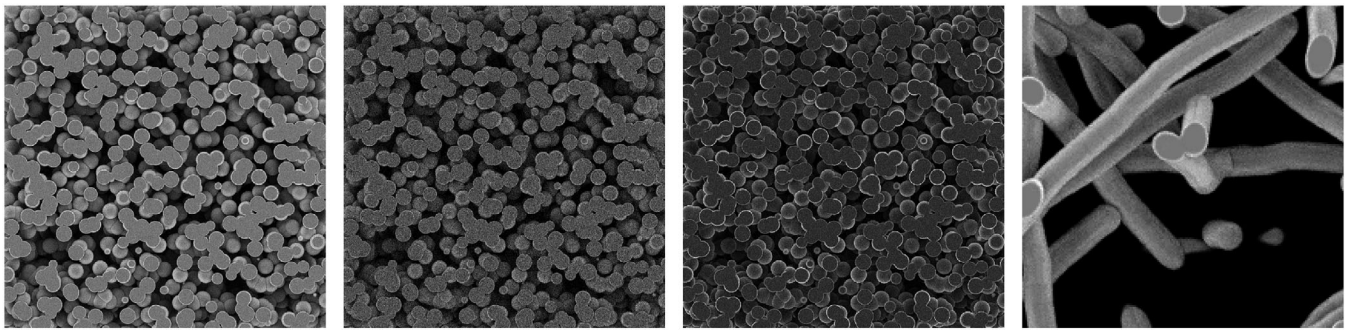


Fig. 7. Simulated scanning electron microscopy images of synthetic structures. First three from left: Boolean model of spheres with sphere radius 15 px, porosity 60%. Left: BoolSpheres – backscattered electron signal, dwelling time 1 μs. Centre left: BoolSpheres0p1μs – backscattered electron signal, dwelling time 0.1 μs. Centre right: BoolSpheresSE – secondary electron signal, dwelling time 1 μs. Right: Altendorf–Jeulin model AJ.

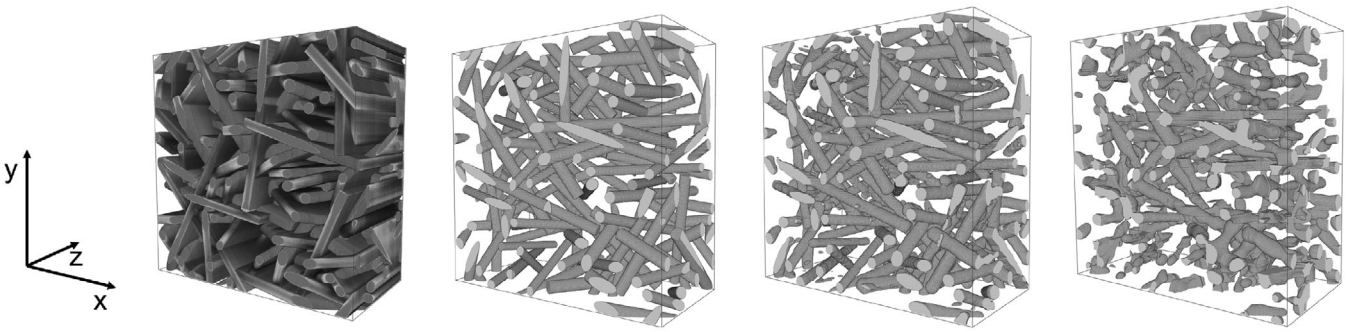


Fig. 8. Results for the random sequential adsorption cylinder packing. From left to right: stack of synthetic backscattered electron images, ground truth, reconstruction result of the trained convolutional neural network with accuracy of the segmentation dice coefficient 0.96, false negative rate 0.005 and false positive rate 0.01, and reconstruction result by the network trained without data augmentation from Section 4, Table 2 with dice coefficient 0.77.

training data (see Fig. 7). This is obviously a result of the low dwelling time and the other SEM mode, respectively. In particular, the grey values strongly influence how well the shine through artefacts are corrected. If the grey value distribution differs notably from the one in the training data, the network tends to wrongly identify shine through artefacts as foreground. This results in artificial elongation of the structure in z-direction whereas z should not be particularly emphasized in realizations of isotropic stochastic geometry models. Thus, elongation in z-direction observed in the reconstruction of an isotropic structure is an indicator for a bad reconstruction.

Rough indicators for artificial anisotropy are the mean chord lengths in x- and z-directions, $\bar{\ell}_x$ and $\bar{\ell}_z$. These are the expected lengths of the segments cut by a random line in x- or z-direction from the solid structure. All models considered here except the random sequential adsorption cylinder packing are isotropic by design. Thus, $\bar{\ell}_x = \bar{\ell}_z$ is expected for a correct reconstruction. Hence, deviation of the quotient $\bar{\ell}_x/\bar{\ell}_z$ from 1.0 reveals artificial anisotropy.

Note that the orientations of the cylinders in the random sequential adsorption packing RSACyl (Fig. 8) follow a distribution assigning slightly more mass to the z-direction. Consequently, the realization features a quotient $\bar{\ell}_x/\bar{\ell}_z = 0.9$ in the direct discretization. Thus, the value of 0.8 for the reconstruction does not necessarily indicate artificial anisotropy.

The reconstruction of the nonstationary Boolean model of spheres BoolSpheresNon shows clearly that the trained network can correctly identify the changes in the volume fraction along the y-axis (Fig. 9). Consequently, the application of a network trained on stationary and isotropic realizations can be extended to new data sets which can be anisotropic, nonstationary or both.

Real data sets

We applied the very same network that trained on one realization of the Boolean cylinder model BoolCyl from Table 1 as described above to a real FIB-SEM data set of a porous ZrO_2 sample. The sample is prepared by spin coating based on a nanocrystalline ZrO_2 powder, sputtered with a gold layer to avoid charging artefacts, and imaged using Carl Zeiss' Crossbeam NVision 40 Field Emission Scanning Electron Microscope. The ZEISS Atlas 5 3D Tomography package allowed for automatic slicing and imaging. For more technical details, we

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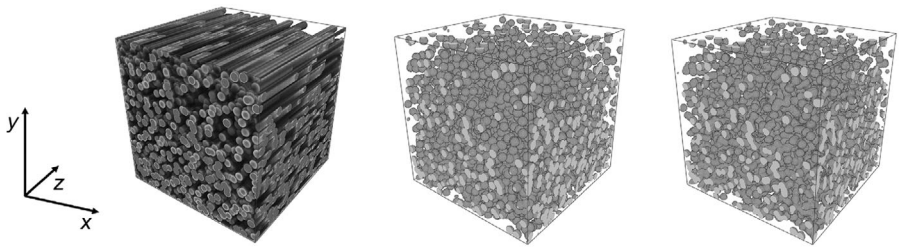


Fig. 9. Results for the nonstationary Boolean model of spheres realization BoolSpheresNon. Left: stack of synthetic backscattered electron images. Centre: ground truth. Right: reconstruction result of the trained convolutional neural networks. The accuracy of the segmentation is dice coefficient 0.97, false negative rate 0.06 and false positive rate 0.002.

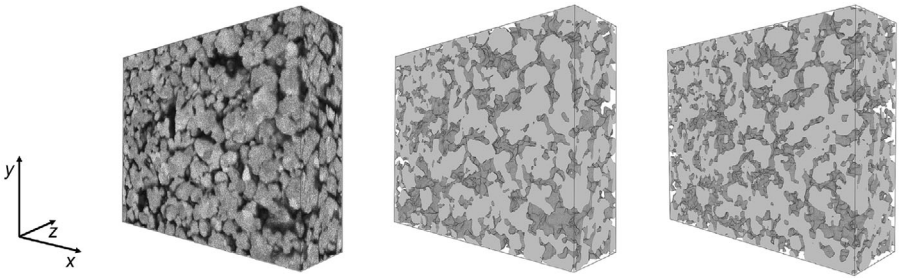


Fig. 10. Results for the nanoporous ZrO_2 sample. Left: stack of backscattered electron images. Centre: reference reconstruction – secondary electron and backscattered electron signals combined using reconstruction by dilation as described by Prill *et al.* (2020); Ott *et al.* (2019). Right: reconstruction result of the trained convolutional neural network. The accuracy of the segmentation is dice coefficient 0.88, false negative rate 0.14 and false positive rate 0.12.

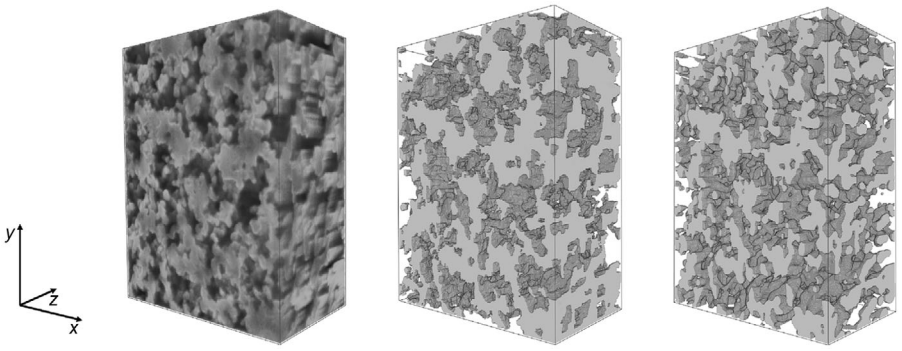


Fig. 11. Results for the nanoporous carbon structure of Wieser *et al.* (2015). Left: stack of backscattered electron images. Centre: reference reconstruction – by Prill's morphological method from Prill *et al.* (2013). Right: reconstruction result of the trained convolutional neural network. The accuracy of the segmentation is dice coefficient 0.88, false negative rate 0.06 and false positive rate 0.12.

refer to Prill *et al.* (2020). The finest layer of the nanoporous membrane analysed there was treated exactly the same way as the ZrO_2 sample at hand.

Both secondary electron and backscattered electron signals are available. Thus reconstruction of the 3D solid structure can rely on morphological combination of the two signals by reconstruction by dilation as described by Prill *et al.* (2020) and Ott *et al.* (2019). Lacking a proper ground truth, we use this reconstruction as reference. The CNN reconstructs the structure based on the BSE signal only. The results are shown in Figure 10. The results are by far not as precise as those for the synthetic data sets. However, the real data were not used in the training at all, the ground truth is not as reliable as the ground truth for the synthetic data sets and the visible deviations are close to the boundary of the structure. This example proves the high potential of the suggested reconstruction strategy as the CNN reaches a reconstruction as good as the one achieved by the signals combination. The latter is rather easy to apply. It relies, however, on the rather exceptional case of both SE and BSE stacks being available and usable, and the SE images being nearly free of shine through artefacts.

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As a second real FIB-SEM data set, we reconsider the nanoporous carbon structure morphologically segmented by Prill *et al.* (2013) and applied in a multiscale simulation by Wieser *et al.* (2015) to study the effect of this additive on the charging and discharging behaviour of a lithium ion battery electrode. The FIB-SEM image was generated using the Helios Nano Lab 600 system of Field Electron and Ion Company. Here, the SE signal was used.

The carbon having a rather rough surface, we trained the network on the Cox–Boolean model 15/20 while leaving all parameters controlling the network setup and training the same as described above.

We do not have a literal ground truth in this case. Instead, the accuracy we report here measures deviation from the surely also not perfect morphological reconstruction by Prill *et al.* (2013). The accuracy achieved by our CNN is nevertheless very promising if we recall the findings of Salzer *et al.* (2015), that a rate of approximately 90 % correctly assigned voxels is the best that can be expected from classical segmentation algorithms. See Figure 11 for a visualization of the result.

Conclusions

We demonstrated, that a modified U-net 3D combined with dedicated augmentation and weighting, can reconstruct synthetic FIB-SEM images of highly porous structures with very high accuracy. In particular, our trained CNN is able to correct the shine through artefacts notorious in FIB-SEM images of highly porous structures.

The achievement much more relevant for practical use is, however, that CNNs trained solely on synthetic images are able to reconstruct real porous structures from FIB-SEM image stacks with very high accuracy. Reconstruction by a CNN trained on synthetic data can readily be generalized to other structures. The only prerequisite is that a geometrically sufficiently similar model can be generated.

A CNN that can deal with all structures in a certain porosity range similarly well would be even better than training the CNN from scratch for each new structure. The next step towards such a generally applicable CNN is to train the network on several geometry models in order to widen its experience with respect to possible local geometric features.

Currently, a couple of parameters have to be chosen during prediction. Most prominently, the grey values in the image and the training data have to be adjusted and the size of the patches has to be adapted to the local structure thickness. Here, the patch size was chosen motivated by the available hardware and grey value histograms were adjusted *ad hoc*. A strategy for choosing and parametrizing these preprocessing steps based on objective criteria would surely improve applicability.

Finally, all images considered here are free of curtaining, charging artefacts and misalignment. It is of course desirable to reconstruct directly from raw FIB-SEM stacks possibly being affected by these three flaws. However, this requires the synthetic FIB-SEM stacks to feature all these artefacts, too.

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